

Blindness to inconsistent local signals in motion transparency from oscillating dots

Ryota Kanai ^{a,*}, Chris L.E. Paffen ^a, Walter Gerbino ^b, Frans A.J. Verstraten ^a

^a *Psychonomics Division, Universiteit Utrecht, Helmholtz Research Institute, Heidelberglaan 2, 3584 CS Utrecht, The Netherlands*

^b *Department of Psychology, University of Trieste, via Sant'Anastasio 12 I-34134 Trieste, Italy*

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Abstract

Visual processing involves hierarchical stages in which local features are initially analyzed and subsequently grouped into objects and surfaces. In the domain of motion perception, transparent motion has been used as a powerful tool to investigate the mechanisms underlying the grouping of local features. Here, we report a novel way of creating motion transparency from oscillating dots (MTOD). In this stimulus, individual dots move back and forth over a small distance. When the dots are oscillating in synchrony, global surfaces are also perceived as moving back and forth. However, when the oscillation desynchronizes, the percept turns into two moving surfaces that are sliding over each other continuously (streaming motion). The percept of MTOD is similar to conventional transparent motion, where individual dots move only in one direction. Also, when streaming motion is perceived, the detection of oscillation is impaired. This blindness to the oscillation becomes stronger, as the signal strength for the streaming motion is increased. These findings suggest that when global visual representations are constructed, weak and inconsistent local signals are discarded.

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1. Introduction

In general, the visual system analyzes a scene using hierarchical processing stages. Initially, local features are analyzed in parallel over the entire visual field. At this stage, local features have not been integrated into objects or surfaces yet. As such these components are perceptually meaningless. A visual environment, however, consists of objects and surfaces rather than local features. To obtain a behaviorally meaningful representation of the visual environment, the visual system needs to combine local features. This integration stage has been extensively studied in the domain of visual motion processing (e.g. Braddick, 1993 for a review).

Visual motion is known to undergo at least two distinct processing stages. At the first stage, motion analysis is performed by local motion filters selective for spatio-temporal orientation (e.g. Adelson & Bergen, 1985; van Santen & Sperling, 1985). This process is

generally thought to occur at the level of the primary visual cortex, also known as the area V1. In the subsequent global stage, these local motions are combined in order to construct the visual environment via integration and segregation. That is, integration by combining the signals arising from a common visual object (or surface), and segregation for those arising from different objects (or surfaces). This process—generally thought to occur at the level of the middle temporal area (MT/V5)—becomes computationally challenging especially under a condition known as motion transparency, in which signals arising from different surfaces occupy the same location in visual space (e.g. Qian, Andersen, & Adelson, 1994; Snowden & Verstraten, 1999; Treue, Hol, & Rauber, 2000; Verstraten, Fredericksen, & van de Grind, 1994). Because of this overlap, these segmentation processes (integration and segregation) have to be performed based on the characteristics of local signals without relying on positional cues.

In this study, we report a new method to create transparent motion: *motion transparency from oscillating dots* (MTOD). For this specific stimulus, the local oscillating dots move back and forth. That is, the local

* Corresponding author. Tel.: +31-30-253-3372; fax: +31-30-253-4511.

E-mail address: r.kanai@fss.uu.nl (R. Kanai).

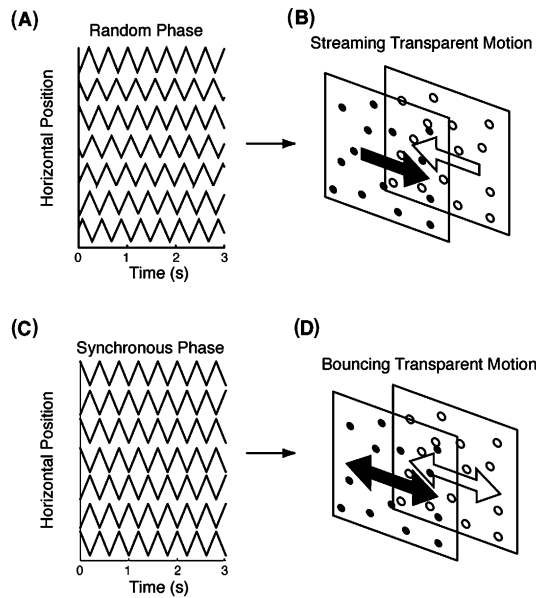


Fig. 1. A schematic illustration of stimuli and typical percepts. (A) Horizontal positions of 7 dots are plotted as a function of time. Each dot oscillates over a small distance. Here, a condition in which the dots have a random initial phase is depicted. (B) The percept for the random-phase condition is illustrated. Two surfaces are perceived as sliding over each other without changing their direction of movement. Filled and open circles are adopted only for purpose of clarity. (C) Horizontal position of 7 dots is depicted. Here, the initial phase of each dot was sampled from either 0 or π . Thus, all the dots change their direction of movement in synchrony. (D) Percept for this stimulus condition is illustrated. Two overlapping surfaces are perceived as bouncing.

dots move back and forth over a short distance at a constant speed, changing their direction of motion periodically (Fig. 1). The percept of MTOD changes depending on the synchronicity of the oscillations. When the directional changes of the dots are asynchronous (Fig. 1A), the result is a compelling percept of streaming motion transparency, which is perceptually similar to conventional transparent motion stimuli. This stimulus configuration is perceived as two surfaces that are sliding over each other continuously without changing their direction of movement, despite the fact that the dots are oscillating locally¹ (Fig. 1B). In contrast, when the dots oscillate in synchrony (Fig. 1C), the percept changes into that of two globally oscillating surfaces, i.e., two transparent surfaces that individually change their directions in phase with oscillations of the local dots (Fig. 1D).

Using a discrimination task, we demonstrate the perceptual similarity between the MTOD and traditional transparent motion. Interestingly, the fact that observers are very poor at discriminating the two types

of transparent motion implies that they are 'blind' to the oscillatory components of the dots. The perceptual presence of the oscillations seems to be overruled by the presence of the current global interpretation. Implications are discussed in detail in Section 5.

2. General methods

2.1. Observers and apparatus

In all experiments, four observers participated. Two are authors on the present paper (RK and CP) and the others were naïve as to the purpose of the experiment. All had normal or corrected-to-normal visual acuity. Stimuli were presented on a 22-in. monitor (LaCie Electron) controlled by a MAC G4 running MATLAB (Mathworks Inc.) and Psychophysics Toolbox (Brainard, 1997; Pelli, 1997). The observers sat in a chair in front of the screen at a viewing distance of 57 cm, with their heads supported by a chinrest.

2.2. Stimuli

The stimuli consisted of 600 oscillatory dots. The position of a dot followed a periodic sawtooth function as following: $X(t) = A \cdot S(2\pi ft + \phi)$, $S(\omega) = 1 - 2\omega/\pi$ ($0 < \omega \leq \pi$), $-3 + 2\omega/\pi$ ($\pi < \omega \leq 2\pi$), where X denotes the horizontal position of a dot, A the amplitude of the oscillation, S a sawtooth periodic function, f the oscillation frequency, t the time elapsed from stimulus onset, and ϕ the initial phase of the oscillation. The oscillation frequency f was set to 2.5 Hz, which means that one full cycle took 400 ms. Oscillation amplitude was 0.4° of visual arc, resulting in a velocity of $4.0^\circ/\text{s}$ either leftwards or rightwards, depending on the phase.

3. Experiment 1

At first glance, the MTOD seems perceptually similar to conventional or traditional motion transparency (hereafter TMT) stimuli. In order to demonstrate this perceptual similarity, we need an objective measure. For this we conducted an experiment in which observers were required to discriminate the MTOD with variable synchronicity from the TMT that consisted of random dots with limited lifetime.

3.1. Methods

3.1.1. Stimuli and procedure

To parametrically shift the degree of oscillation synchronicity, we manipulated the range of distributions from which the initial phases of the dots were sampled (Fig. 2A–C). This way, the synchronicity was varied

¹ To view these moving stimuli, see <http://www.fss.uu.nl/psn/Kanai/transp.html>.

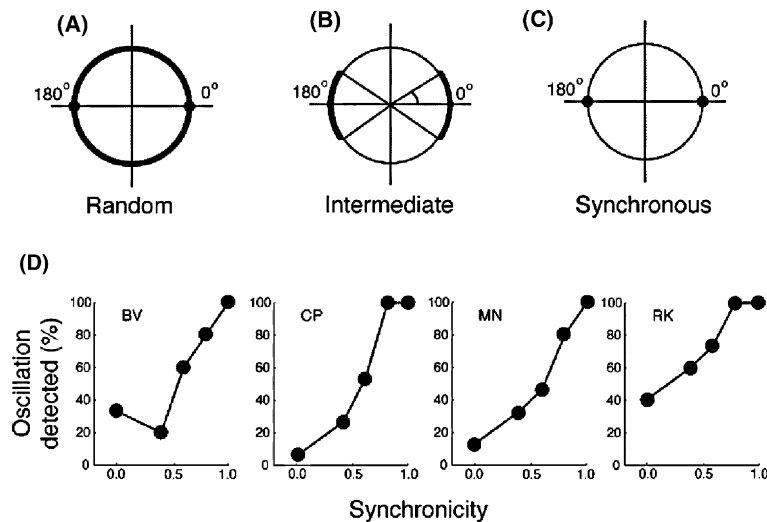


Fig. 2. An illustration of the phase distribution and the results of the experiment in which the synchronicity of oscillation was systematically varied. (A) When the synchronicity was zero, the initial phase of an oscillating dot was uniformly sampled from 360°. (B) A higher rate of synchronicity was obtained by restricting the sampling source (thick line). (C) The maximum synchronicity was obtained when the initial phase was sampled from either of two points that are 180° apart from each other. An index of the synchronicity was defined between 0 and 1 by linearly interpolating the random-phase condition and synchronous-phase condition. (D) The results of four observers. The detection performance is plotted as a function of synchronicity.

gradually between two extreme conditions: The random-phase condition (Fig. 2A) and the synchronous-phase condition (Fig. 2C). The distribution was set symmetrically around a unit circle such that at any time during the stimulus presentation, the balance between leftwards and rightwards motions was kept constant. This manipulation was necessary in order to remove additional cues for observers to detect oscillating dots such as an imbalance between leftwards and rightwards movements. The index of synchronicity was defined between 0 and 1, corresponding to the random-phase condition and synchronous-phase condition, respectively. Intermediate levels of synchronicity were created by adjusting the angle (θ , in Fig. 2B) between 0° and 180°. The synchronicity index (I_s) was defined as $(\pi - \theta)/\pi$. The stimuli consisted of 600 dots. Twenty trials were performed for each synchronicity level. TMT stimuli also consisted of 600 dots with limited lifetime (200 ms) and they were interleaved on half the trials. The lifetime (200 ms) was chosen so that it corresponds to half the cycle of the oscillation (i.e., the duration for which the oscillatory dots continuously move in one direction).

In our pilot observations, it was clear that if the observation duration was sufficiently long, observers were able to discriminate MTOD stimuli from the TMT stimuli by carefully attending to individual dots. To optimize the difficulty of the task, we therefore presented the stimuli only for 800 ms (i.e., 2 oscillation cycles). The dots were shown within a square area of 7.8° by 7.8°, whose upper edge was at a distance of 5.9° below the fixation point.

The observers' task was to detect the presence of oscillating dots in the stimulus. A beep was given as feedback when observers incorrectly reported the presence of oscillatory dots in catch trials.

3.2. Results and discussion

The results for individual observers are shown in Fig. 2C. The detection performance of the oscillating dots increased monotonically as the synchronicity increased (Spearman rank-order correlation, $R_s = 0.9148$, $P < 0.001$). The detection was easy in the synchronous-phase condition, because the percept of bouncing transparent motion was sufficient for observers to report the presence of oscillatory dots (see Fig. 1D). Thus, in the synchronous-phase condition, all observers were able to detect the oscillatory dots in all the trials. The false alarm rate was 0.8%, 0.0%, 26.7%, and 1.9% for observer BV, CP, MN, and RK, respectively. In the conditions with lower synchronicity, the detection of oscillatory dots was more difficult. The poor performance in those conditions indicates the difficulty in discriminating the MTOD and TMT. This is likely due to the similarity of the global percept between the two. In contrast to the synchronous-phase condition, the perceptual appearance of the MTOD (i.e., streaming transparent motion) does not indicate the presence of the oscillatory dots. Therefore, in such a case, the observers needed to isolate individual dots in order to correctly report the presence of oscillatory dots.

Interestingly, the observers were not able to detect the oscillatory dots in the random-phase condition, despite

the fact that all the dots were oscillating and the observers were trying to find them. This implies that they are ‘blind’ to the oscillatory components of the dots.

4. Experiment 2

What caused the *blindness* to the oscillations in MTOD? We can think of two possible explanations. It could be that the signals arising from directional reversals are integrated over too short a time-window. That is, when the oscillations are asynchronous, they are not integrated efficiently. Therefore, the output signals after integration are not strong enough for the system to detect. Consequently, the detection of oscillation is poor at low synchronicity. A second possibility is that the blindness may be due to a conflict between interpretations. The oscillations of the dots result in an incompatible interpretation for the visual system, given the presence of the streaming transparent motion interpretation. An analogous situation can be found in the perception of bistable stimuli. While viewing ambiguous figures like the Necker cube, two mutually incompatible interpretations are latent in the stimulus, yet only one interpretation is perceived. As long as one interpretation is dominant, the other is not consciously perceived. Similarly, in the case of the oscillatory dots, the information inconsistent with the dominant interpretation (the global surfaces) may not be consciously perceived.

In the next experiment, we examine this possibility. To avoid the issue of the time-window discussed above, we keep the oscillations always synchronous (equivalent to the synchronous-phase condition in Experiment 1), while we manipulate the signal strength for the global percept by varying the coherence of the moving dots (see Fig. 3A–C). The rationale of this experiment is that with higher motion coherence, the local oscillations become more incompatible with the global percept, and therefore less visible for observers. Following the same line of reasoning; when the local oscillations are embedded in incoherent noise, they will be less incompatible, and thus more detectable. Thus, the prediction for this experiment is that the blindness to oscillations increases with higher coherence.

4.1. Methods

4.1.1. Stimuli and procedure

Oscillating dots (target; see Fig. 3A) were superimposed on a transparent motion stimulus with a variable signal level (distractor; see Fig. 3B). We varied the coherence in the distractor (5%, 10%, 20% and 90%). These trials (target-present) were interleaved with the trials where the target was replaced with traditional motion transparency (TMT) consisting of 600 random dots. The dots had a lifetime of 200 ms, which corresponds to half a cycle at 2.5 Hz oscillation frequency. The dots were updated at random locations and disappeared after their lifetime. In order to avoid

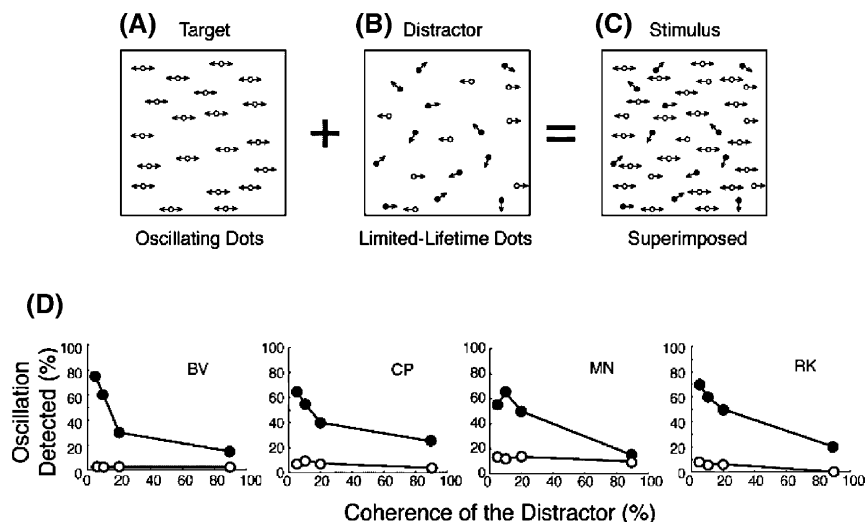


Fig. 3. A schematic illustration of the stimuli used in the second experiment. Transparent motion generated by 300 oscillating dots (target) was superimposed on a distractor stimulus consisting of 300 moving dots with a limited lifetime (200 ms). (A) 300 dots moving back and forth at a frequency of 2.5 Hz with an amplitude of 0.4° . The stimulus was presented for two cycles (i.e., 800 ms). (B) The distractor consisted of 300 random dots with a limited lifetime of 200 ms. The proportion of signal dots was varied between 5%, 10%, 20% and 90%. Here, an example of 40% signal is illustrated. For illustration purposes, noise dots are shown as filled-circles and signal dots by open-circles. In the experiment, all dots were white on a black background. (C) A schematic image of superimposition of a target and a distractor. (D) Results for four observers. The proportion of trials in which observers correctly reported local oscillation is indicated by the solid circles (●), and the false alarm rate by the open circles (○).

synchronous onset or offset, the dots appearing in the first frame of a trial were assigned a random duration (lifetime < 200 ms). Four different levels of coherence were used in the target-absent trials (52.5%, 55%, 60% and 95%), which were equal to the coherence level in the target-present trials. During a trial, the stimulus was presented for 800 ms (i.e., 2 oscillation cycles). Again, the dots were shown within a square area of 7.8° by 7.8° , whose upper edge was at a distance of 5.9° below the fixation point. Each stimulus condition was performed 20 times, resulting in a total of 160 trials.

Observers were asked to indicate the presence or absence of the oscillatory dots. Feedback was given by a beep when observers mistakenly reported the presence in the target-absent trials.

4.2. Results and discussion

The results are shown in Fig. 3D. The detection of oscillatory dots decreased monotonically as the coherence level in the distractor increased (Spearman rank-order correlation, $R_s = -0.920$, $P < 0.001$). Thus, the detection of oscillatory dots critically depended upon the signal strength for global motion. This suggests that when the global percept of streaming motion is supported by high coherence, the blindness to oscillation increased. The results support the idea that the blindness to oscillation is due to its incompatibility with the global percept.

However, a weak activation of the global motion system does not seem to be sufficient. In other words, when streaming transparent motion was barely visible, the detection of the oscillation was not impaired. In all stimulus conditions, an equal amount of oscillatory dots were always moving coherently in the horizontal directions. In fact, streaming transparent motion was visible even in the condition with the lowest coherence level (5%). Thus, the critical factor for the blindness is not the mere presence of global moving surfaces, but the strength of the signal supporting the presence of global surfaces.

5. General discussion

In the present study, we reported a new way of generating transparent motion. When the dots oscillate asynchronously, the MTOD is perceptually very similar to the more classical types of transparent motion. In MTOD, the oscillatory components of motion become invisible for observers. This *blindness effect* is dependent on the signal strength supporting the global surface interpretation. Our experiments suggest that this blindness is due to the incompatibility of their representation with the global percept of streaming motion.

In our MTOD display, the global percept drastically changed depending on the synchronicity of directional changes of the local dots. Such perceptual phenomena have been reported earlier using slightly different displays (Bravo & Watamaniuk, 1995; Gerbino & Berneti, 1984; Treue, Andersen, Ando, & Hildreth, 1995; Watamaniuk, Flinn, & Stohr, 2003). For example, Treue et al. (1995) used a cylinder-shaped structure-from-motion stimulus consisting of the dots that changed the direction of rotation periodically. They showed that observers hardly notice the change of the cylinder rotation, especially when the local dots reversed their rotation direction asynchronously. Bravo and Watamaniuk (1995) used a motion display in which each dot moved with two speeds (slow and fast) alternately, but in the same direction. In this case, asynchronous speed changes resulted in the percept of two superimposed sheets of moving dots. Moreover, Watamaniuk et al. (2003) showed that dots moving in two directions in alternation also result in the percept of transparent motion. These examples show dissociation of the behavior of individual dots and the global percept. Common to all these stimulus types is that the asynchronous alternations in speed or direction is the key to obtain a clear percept of transparent motion without being disturbed by the changes in the local dots.

The blindness to oscillations in MTOD is perceptually similar to a visual phenomenon known as *motion capture* (e.g. Ramachandran & Cavanagh, 1987). As is true for our stimulus, in motion capture the individual motion signals, which are certainly present in the stimulus configuration, are not perceived. Thus for both phenomena, incompatible local signals are not perceived in the presence of coherent global motion. In Experiment 2, we showed that the coherence for global motion is the critical factor for the decrease in performance in oscillation detection. This is in line with previous findings indicating that the more compelling the percept of global motion is, the stronger the effect of motion capture becomes (Ramachandran & Anstis, 1983; Yo & Wilson, 1992).

How does the blindness arise? One possible account is that local signals can be perceived only if they are part of a global representation. However, this cannot fully explain our results. In Experiment 2, global motion was perceived in all conditions. However, the effect appeared mainly in those conditions where the signal coherence was high (e.g. 90%). Thus, the critical factor for the blindness to oscillation is not the mere presence of global moving surfaces, but *strong* signals for the global representation.

An alternative account for the blindness effect is that local signals are suppressed by a strong global representation before reaching visual awareness. Mutually suppressive interactions between simultaneously present interpretations are known to exist in higher visual

cortical areas (Desimone, 1998; Desimone & Duncan, 1995; Kastner & Ungerleider, 2000). It is plausible that such an inhibitory interaction exist for early visual processing as well. If this were the case, then the inhibitory effect can be accounted for by suppression of the local signals in the early visual processing area such as V1 via feedback from the higher visual area representing the global surfaces (e.g. area MT). This is consistent with our finding that the blindness effect critically depends on the strength of the signals supporting a global representation. Strong signals for the global surfaces activate the higher visual areas. This may result in a stronger suppression of inconsistent signals through feedback. In addition, such a suppressive mechanism for inconsistent local signals would be advantageous for the visual system, because it facilitates the construction of surface representation in the presence of conflicting information or noise.

The idea of the suppression dependent on the signal level for global motion is applicable to other visual phenomena. For example, when we view ambiguous stimuli like a Necker cube, the percept alternates between possible interpretations. The possible interpretations are not perceived simultaneously. This means that the information that is inconsistent with the dominant percept is suppressed and does not reach visual awareness. In a similar vein, motion-induced blindness (Bonneh, Cooperman, & Sagi, 2001) may be explained by the suppression of inconsistent signals. For this phenomenon, salient dots placed on a field of coherently moving dots disappear from our awareness despite constant stimulation of the retina. Those salient dots are incompatible with the representation of global motion (e.g. structure-from-motion). As a result of this incompatibility, the signals for the dots are suppressed, leading to their perceptual disappearance.

So far it is not known whether such inhibition is actually involved in early visual processing, and our present data cannot prove or disprove this point. However, this is an interesting empirical question and as such further studies can address this question.

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References

- Adelson, E. H., & Bergen, J. R. (1985). Spatiotemporal energy models for the perception of motion. *Journal of the Optical Society of America A*, 2, 289–299.
- Bonneh, Y., Cooperman, A., & Sagi, D. (2001). Motion induced blindness in normal observers. *Nature*, 411, 798–801.
- Braddick, O. (1993). Segmentation versus integration in visual motion processing. *Trends in Neuroscience*, 16, 263–269.
- Brainard, D. H. (1997). The Psychophysics Toolbox. *Spatial Vision*, 10, 433–436.
- Bravo, M. J., & Watamaniuk, S. N. J. (1995). Evidence for two speed signals: a coarse local signal for segregation and a precise global signal for discrimination. *Vision Research*, 35, 1691–1697.
- Desimone, R. (1998). Visual attention mediated by biased competition in extrastriate visual cortex. *Philosophical Transactions of the Royal Society of London Series B*, 353, 1245–1255.
- Desimone, R., & Duncan, J. (1995). Neural mechanisms of selective visual attention. *Annual Review of Neuroscience*, 18, 193–222.
- Gerbino, W., & Berneti, L. (1984). *Perception*, 13, A38–A39.
- Kastner, S., & Ungerleider, L. G. (2000). Mechanisms of visual attention in the human cortex. *Annual Review of Neuroscience*, 23, 315–341.
- Pelli, D. G. (1997). The VideoToolbox software for visual psychophysics: transforming numbers into movies. *Spatial Vision*, 10, 437–442.
- Qian, N., Andersen, R. A., & Adelson, E. H. (1994). Transparent motion perception as detection of unbalanced motion signals: I. Psychophysics. *Journal of Neuroscience*, 14, 7357–7366.
- Ramachandran, V. S., & Anstis, S. M. (1983). Displacement thresholds for coherent apparent motion in random-dot patterns. *Vision Research*, 23, 1719–1724.
- Ramachandran, V. S., & Cavanagh, P. (1987). Motion capture anisotropy. *Vision Research*, 27, 97–106.
- Snowden, R. J., & Verstraten, F. A. J. (1999). Motion transparency: making models of motion perception transparent. *Trends in Cognitive Sciences*, 3, 369–377.
- Treue, S., Andersen, R. A., Ando, H., & Hildreth, E. C. (1995). *Vision Research*, 35, 139–148.
- Treue, S., Hol, K., & Rauber, H. J. (2000). Seeing multiple directions of motion—physiology and psychophysics. *Nature Neuroscience*, 3, 270–276.
- van Santen, J. P. H., & Sperling, G. (1985). Elaborated Reichardt detectors. *Journal of the Optical Society of America A*, 2, 300–321.
- Verstraten, F. A. J., Fredericksen, R. E., & van de Grind, W. A. (1994). The movement aftereffect of bi-vectorial transparent motion. *Vision Research*, 34, 349–358.
- Watamaniuk, S. N. J., Flinn, J., & Stohr, R. E. (2003). Segregation from direction differences in dynamic random-dot stimuli. *Vision Research*, 43, 171–180.
- Yo, C., & Wilson, H. R. (1992). Moving two-dimensional patterns can capture the perceived directions of lower or higher spatial-frequency gratings. *Vision Research*, 32, 1263–1269.